

## Computer Graphics – June 24<sup>th</sup>, 2020

Name: \_\_\_\_\_

Seat Number: \_\_\_\_\_

Matriculation Number: \_\_\_\_\_

Please read the following instructions carefully:

- You have sixty (60) minutes to complete the exam.
- You can answer in German or in English.
- Check that the exam is complete (13 pages on 7 sheets) and readable!
- Please use a blue or black ballpen, not a red or green one. Do not use a pencil!
- Only text on the stapled sheets will be evaluated.
- If needed, we staple additional blank sheets.
- Only writing material is allowed!

I have read and understood the instructions above.

Erlangen, June 24<sup>th</sup>, 2020

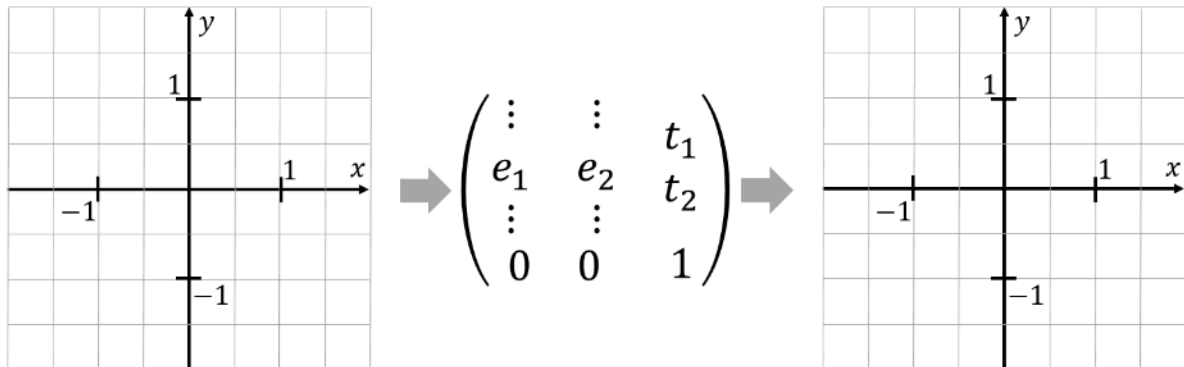
\_\_\_\_\_  
Signature

| 1  | 2  | 3  | 4  | 5  | ALL |
|----|----|----|----|----|-----|
| 12 | 12 | 10 | 16 | 10 | 60  |

Grade:

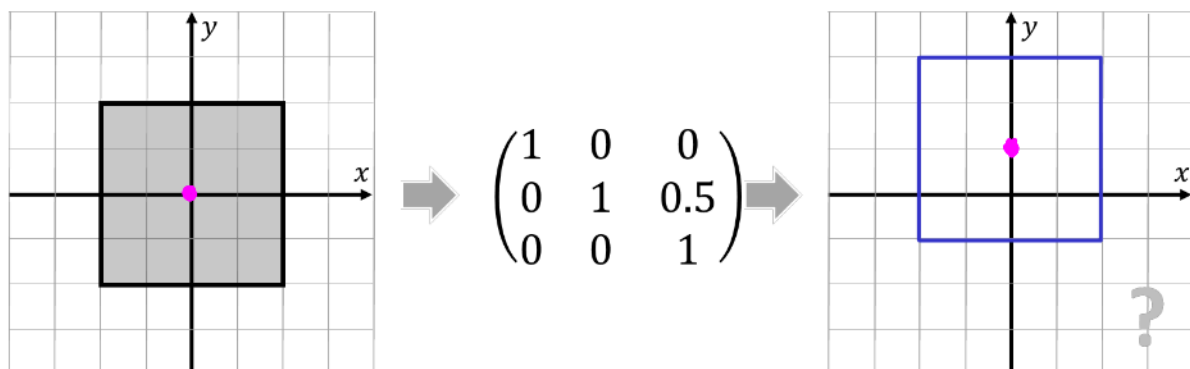
## Assignment 1: Affine Mappings in 2D [12 points]

Affine mappings in 2D can be expressed using homogeneous coordinates and  $3 \times 3$  matrices with the following structure, where  $(\vec{e}_1, \vec{e}_2)$  is the linear part (image of base vectors) and  $(t_x, t_y)$  is the translation:

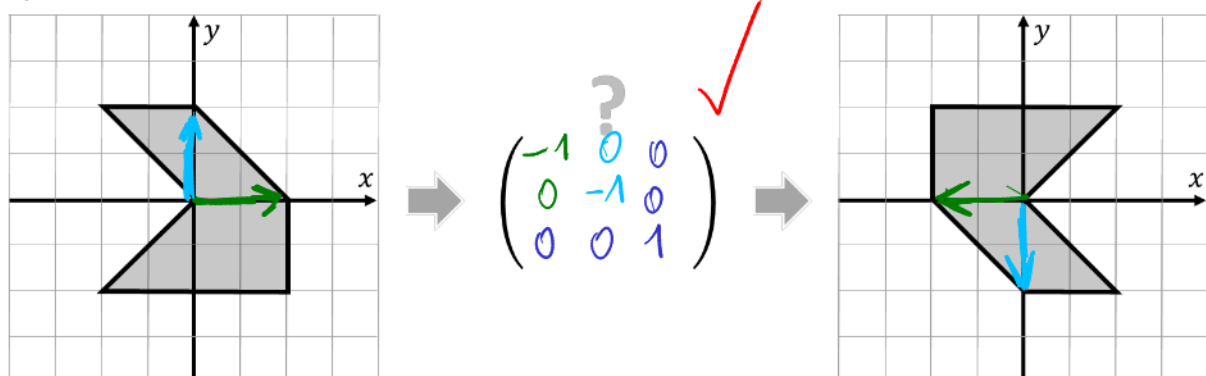


Each of the sub-assignments a) to e) deals with one mapping (in the coordinate system depicted above) and consists of two images (before and after mapping) and one matrix that has the desired effect. Complete each of the sub-assignments by filling in the missing part marked with a question mark.

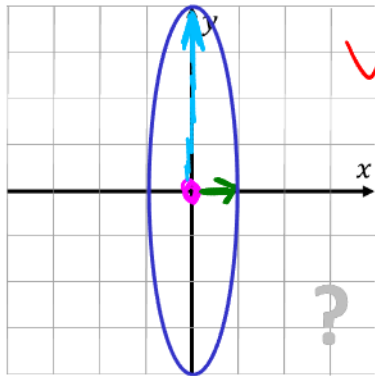
a)



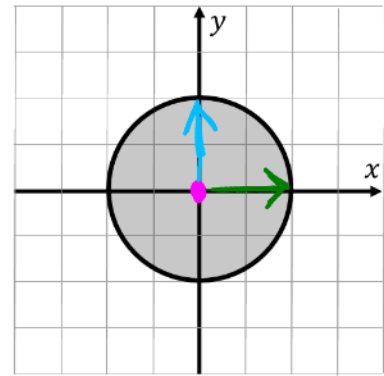
b)



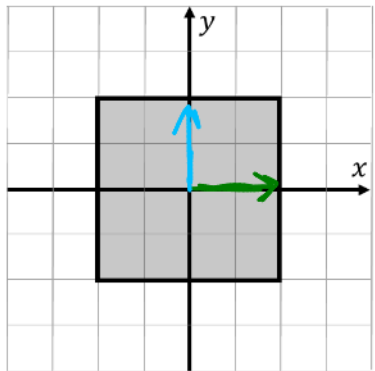
c)



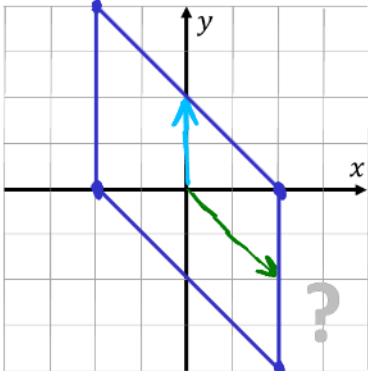
$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$



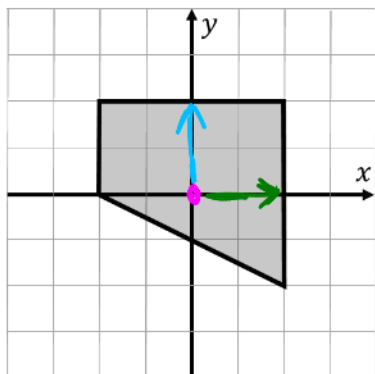
d)



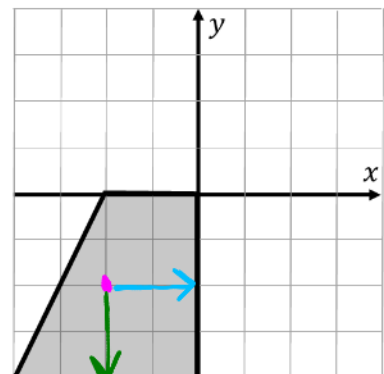
$$\begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$



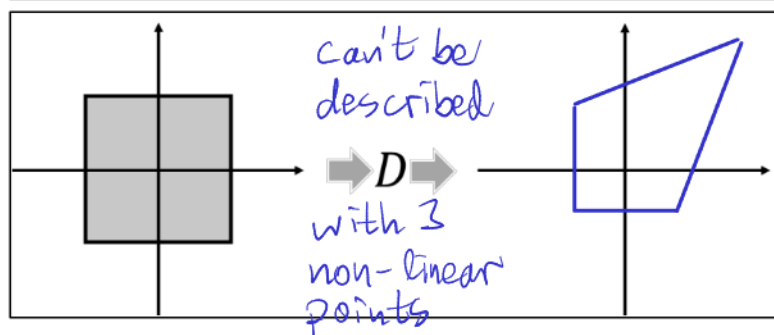
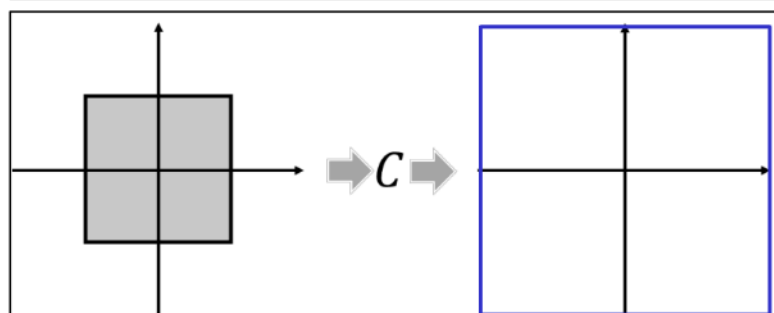
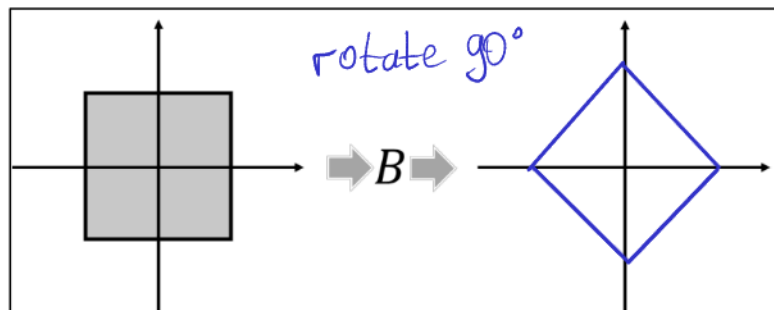
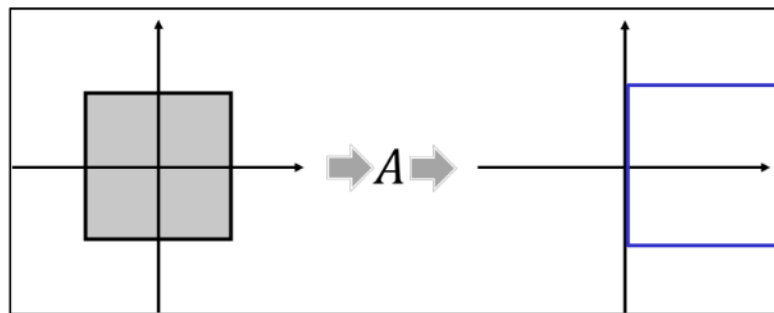
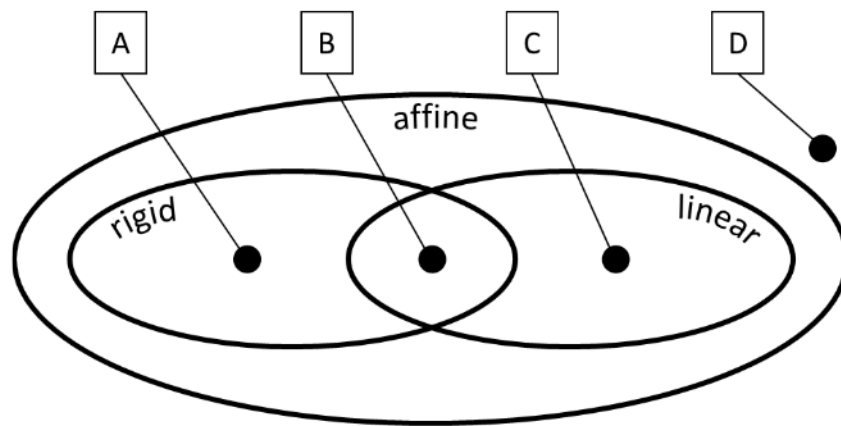
e)



$$\begin{pmatrix} 0 & 1 & -1 \\ -1 & 0 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$



f) Below you see the set diagram of rigid, linear and affine mappings known from the lecture. A, B, C and D stand for four mappings, each in one subset of the diagram. In the boxes below the diagram, sketch results of A, B, C and D when applied to a square – there is **more than one solution!**



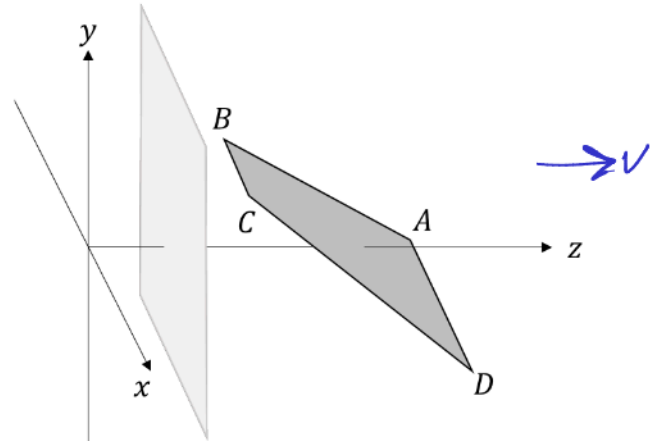
## Assignment 2: Perspective Projection &amp; Texture Mapping [12 points]

Given is a trapezoid  $ABCD$  with the four vertices

$$A = (-1, -1, 4), B = (-0.5, 0.5, 2), \\ C = (0.5, 0.5, 2), D = (1, -1, 4).$$

We project the rectangle perspectively (with the  $z$ -axis as view direction) using the following homogeneous matrix:

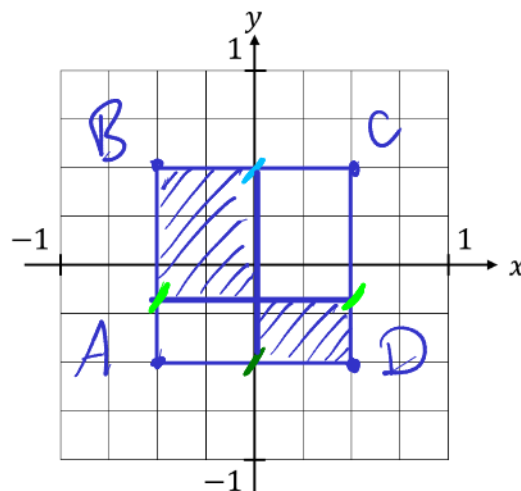
$$M = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$



- a) Map the vertices to image space using matrix  $M$  and fill in the table with the mapping results. For all vertices, give both their two-dimensional image coordinates and their depth value after normalization!

| Vertex | Image coordinates $\in [-1, 1]^2$ | Depth after normalization $\in [-1, 1]$ |
|--------|-----------------------------------|-----------------------------------------|
| A      | $-1/2, -1/2$                      | $-3/4$                                  |
| B      | $-1/2, 1/2$                       | $-1/2$                                  |
| C      | $1/2, 1/2$                        | $-1/2$                                  |
| D      | $1/2, -1/2$                       | $-3/4$                                  |

- b) Sketch the resulting shape in image space into the following 8x8 pixel grid:



2  
0

$$\tilde{A} = \begin{pmatrix} A_1 \\ A_2 \\ A_3 \\ 1 \end{pmatrix} \quad \text{"homogenized"}$$

$$M\tilde{A} = \begin{pmatrix} -2 \\ -2 \\ -3 \\ 4 \end{pmatrix} \xrightarrow{\text{perspective divide}} \begin{pmatrix} -1/2 \\ -1/2 \\ -3/4 \end{pmatrix} \begin{matrix} x \\ y \\ z \end{matrix} \left. \vphantom{\begin{pmatrix} -1/2 \\ -1/2 \\ -3/4 \end{pmatrix}} \right\} \begin{matrix} \text{image-space} \\ \text{depth} \end{matrix}$$

$\nwarrow w=4$

$$M\tilde{B} = \begin{pmatrix} -1 \\ 1 \\ -1 \\ 2 \end{pmatrix} \rightarrow \begin{pmatrix} -1/2 \\ 1/2 \\ -1/2 \end{pmatrix}$$

$$M\tilde{C} = \begin{pmatrix} 1 \\ 1 \\ -1 \\ 2 \end{pmatrix} \rightarrow \begin{pmatrix} 1/2 \\ 1/2 \\ -1/2 \end{pmatrix}$$

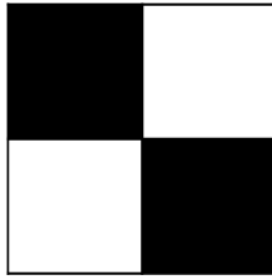
$$M\tilde{D} = \begin{pmatrix} 2 \\ -2 \\ -3 \\ 4 \end{pmatrix} \rightarrow \begin{pmatrix} 1/2 \\ -1/2 \\ -3/4 \end{pmatrix}$$

$$M \frac{1}{2}(\tilde{A} + \tilde{B}) = M \begin{pmatrix} -3/4 \\ -1/4 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} -3/2 \\ -1/2 \\ -5 \\ 3 \end{pmatrix} \rightarrow \begin{pmatrix} -1/2 \\ -1/3 \\ -5/3 \end{pmatrix} \quad \}$$

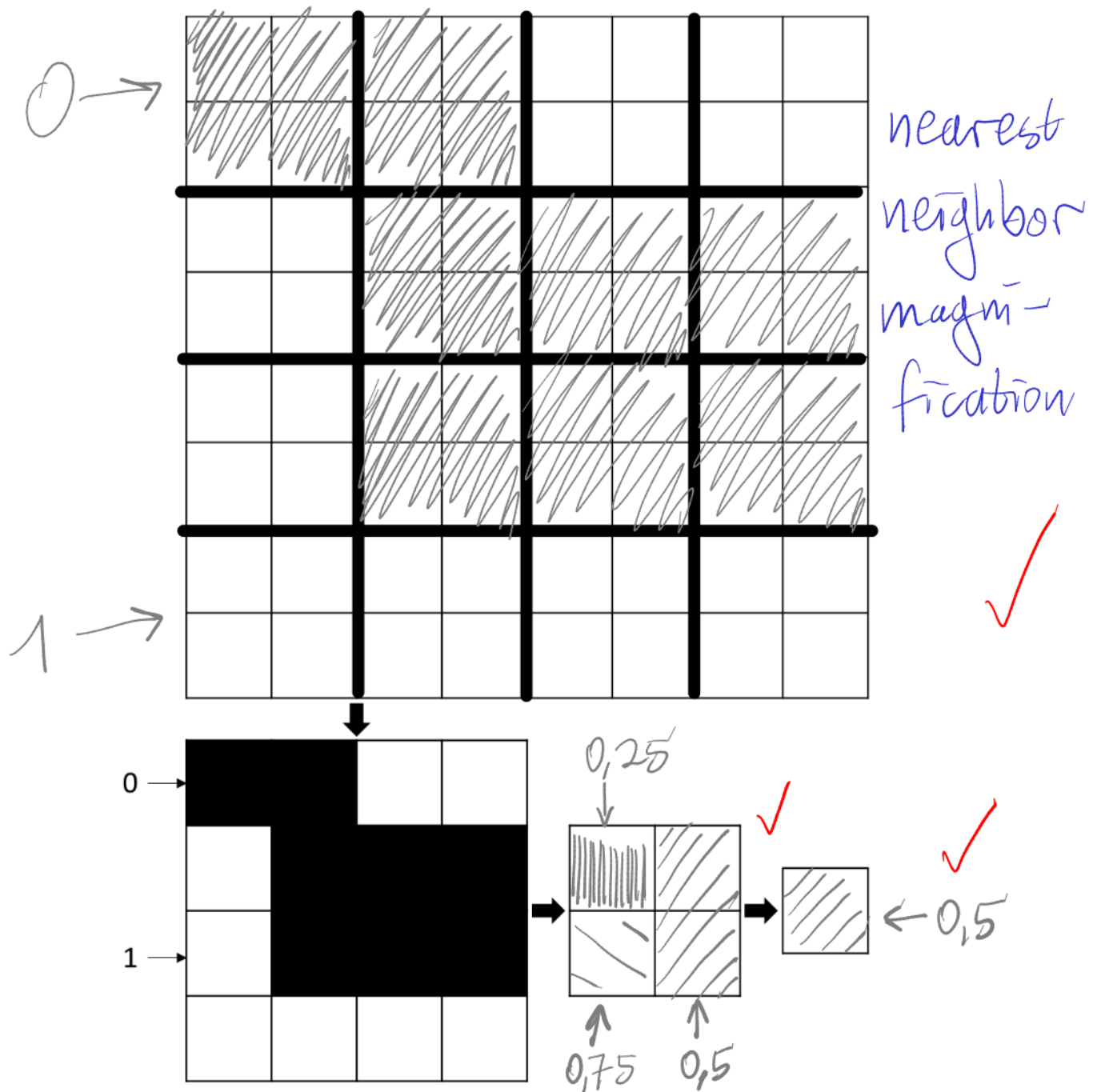
$$M \frac{1}{2}(\tilde{B} + \tilde{C}) = M \begin{pmatrix} 0 \\ 1/2 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ -1 \\ 2 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 1/2 \\ -1/2 \end{pmatrix} \quad \}$$

$$M \frac{1}{2}(\tilde{A} + \tilde{D}) = M \begin{pmatrix} 0 \\ -1 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -2 \\ -3 \\ 4 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ -1/2 \\ -3/4 \end{pmatrix} \quad \}$$

- c) Next, the following 2x2 texture should be mapped to the trapezoid in a perspective correct manner. Add a sketch of this mapped texture to your sketch in b).



- d) Below you see a greyscale texture of a duck (roughly). Compute the missing levels of the MIP map pyramid this texture belongs to. Label your sketch with values between 0 (black) and 1 (white) as it is done for the given texture.



## Assignment 3: Line Rasterization [10 points]

In the lecture, you learned about the Bresenham algorithm for line rasterization:

```
function line(int x0, int y0, int x1, int y1)
{
    int x = x0, y = y0
    int  $\Delta x$  = x1 - x0,  $\Delta y$  = y1 - y0
    int D =  $\Delta x - 2\Delta y$ 
    int  $\Delta DE = -2\Delta y$ ,  $\Delta DNE = 2(\Delta x - \Delta y)$ 
    while (x <= x1) {
        draw_pixel(x,y)
        x = x + 1
        if (D < 0) {
            y = y + 1
            D = D +  $\Delta DNE$ 
        }
        else
            D = D +  $\Delta DE$ 
    }
}
```

- a) Rasterize the line from (1,1) to (9,4) with this algorithm. Use the table below and fill in the variable values *at the beginning* of each loop iteration. Mark the set pixels in the grid below.

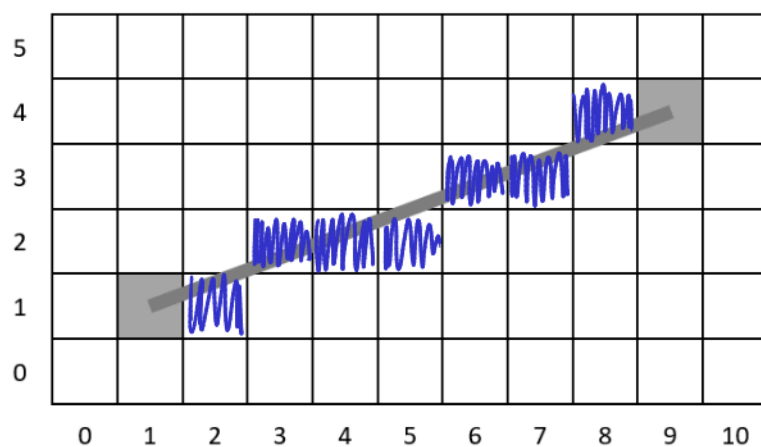
$$\Delta x = 8, \Delta y = 3$$

$$\Delta DE = -6$$

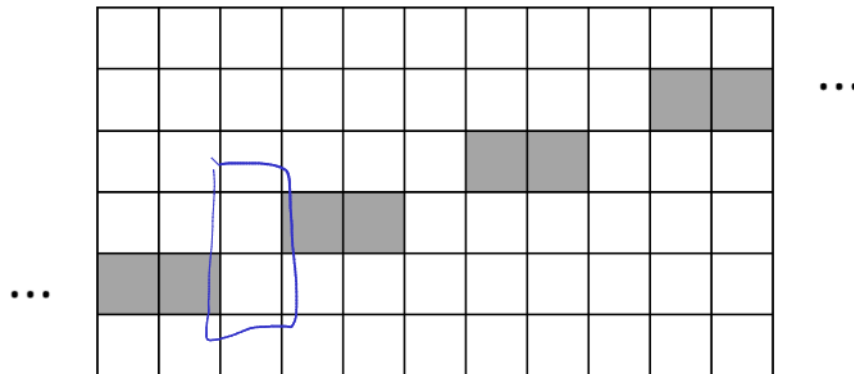
$$\Delta DNE = 10$$



| x | y | D  |
|---|---|----|
| 1 | 1 | 2  |
| 2 | 1 | -4 |
| 3 | 2 | 6  |
| 4 | 2 | 0  |
| 5 | 2 | -6 |
| 6 | 3 | 4  |
| 7 | 3 | -2 |
| 8 | 4 | 8  |
| 9 | 4 | 2  |

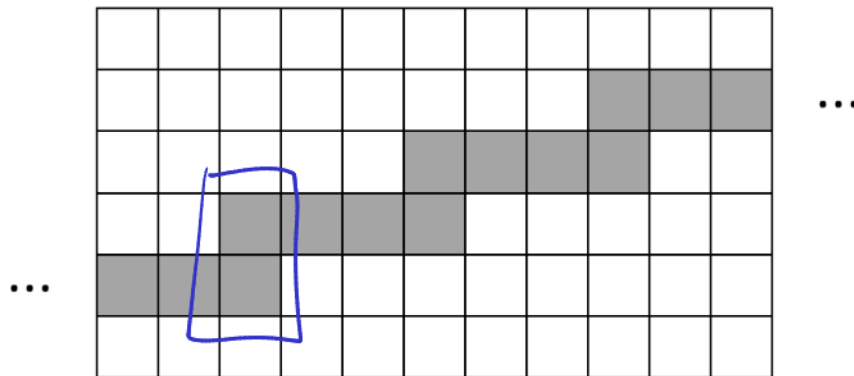


- b) For each of the following lines, briefly state why it cannot be a result of the Bresenham algorithm. Refer back to the algorithm for your explanation.

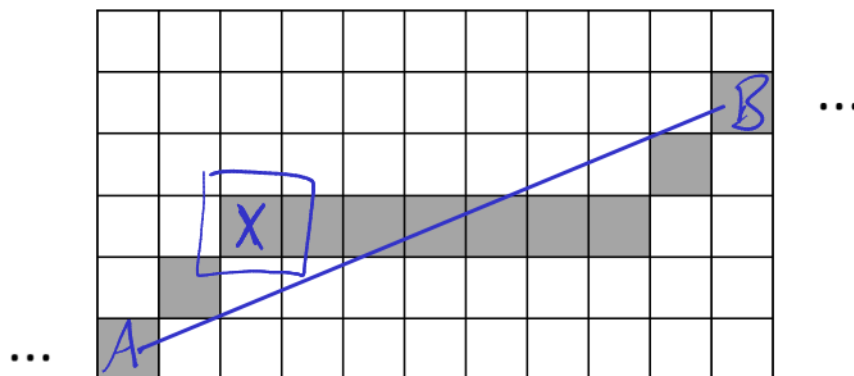


In the loop the current pixel is drawn and  $x$  is incremented by  $+1$ . Therefore there can't be columns without any filled pixel in between the start and end point of a line.





x is always incremented by +1 in the loop.  
As a result all columns between start and end point have exactly one filled pixel.

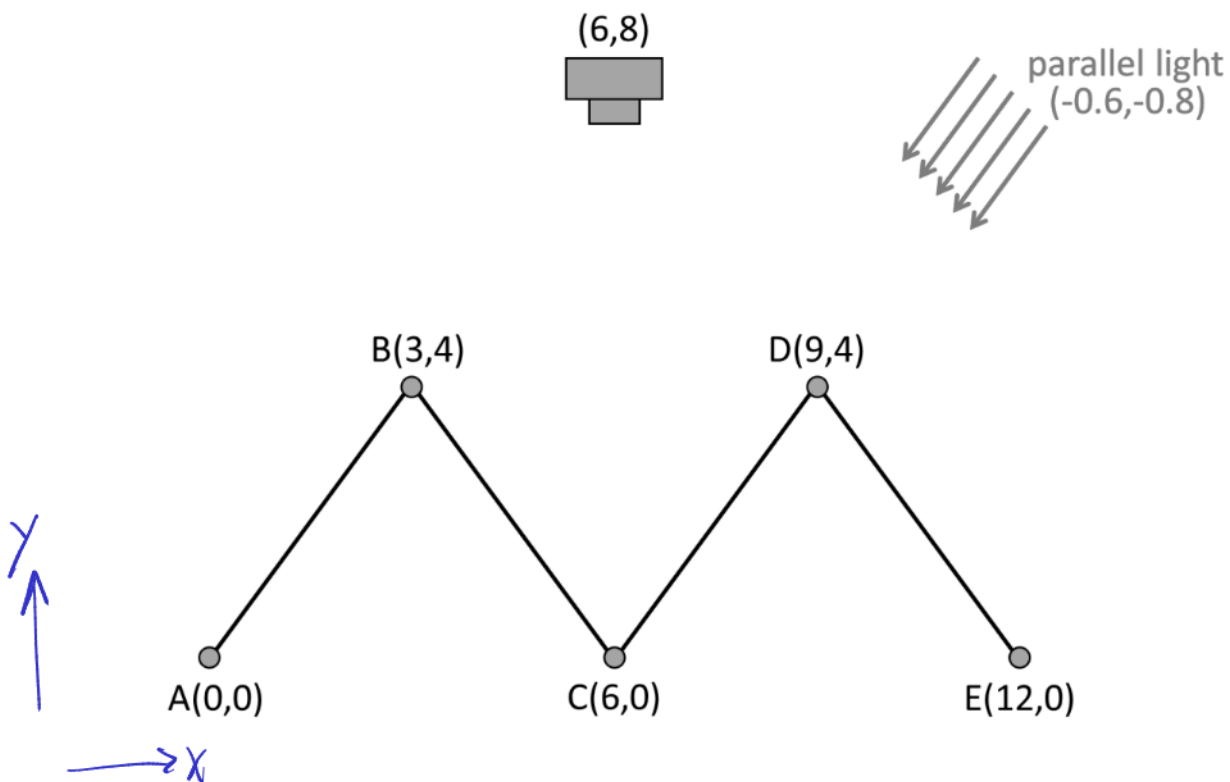


The filled pixels are not on a line:  
Pixel X does not touch the line between the first pixel A and the last pixel B at all.



## Assignment 4: Phong Lighting & Shading [16 points]

The image below shows a surface made up of multiple facets. The surface is lit by a parallel light source with rays in direction  $(-0.6, -0.8)$  and captured by a camera located at  $(6, 8)$ .



- a) For the two points B and D, write down the respective normal vector  $n$ , light vector  $l$ , view vector  $v$  and reflection vector  $r$ ! Hint: The normal of a point can be computed as the average normal of the adjoining faces (or lines, in 2D).

|     | B                                                                                                                                 | D                                           |
|-----|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| $n$ | $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$                                                                                            | $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$      |
| $l$ | $\begin{pmatrix} 0,6 \\ 0,8 \end{pmatrix}$                                                                                        | $\begin{pmatrix} 0,6 \\ 0,8 \end{pmatrix}$  |
| $v$ | $\begin{pmatrix} 3/5 \\ 4/5 \end{pmatrix}$                                                                                        | $\begin{pmatrix} -3/5 \\ 4/5 \end{pmatrix}$ |
| $r$ | $\begin{pmatrix} -0,6 \\ -0,8 \end{pmatrix} + \begin{pmatrix} 0 \\ 1,6 \end{pmatrix} = \begin{pmatrix} -0,6 \\ 0,8 \end{pmatrix}$ | $\begin{pmatrix} -0,6 \\ 0,8 \end{pmatrix}$ |

$$l \cdot n$$

$$0,8$$

$$0,8$$

$$r = -l + 2(l \cdot n)n$$

$$\begin{pmatrix} 6 \\ 8 \end{pmatrix} - \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \end{pmatrix} \leadsto v_A = \frac{\begin{pmatrix} 3 \\ 4 \end{pmatrix}}{\left| \begin{pmatrix} 3 \\ 4 \end{pmatrix} \right|} = \frac{1}{\sqrt{25}} \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 3/5 \\ 4/5 \end{pmatrix}$$

$$\begin{pmatrix} 6 \\ 8 \end{pmatrix} - \begin{pmatrix} 9 \\ 4 \end{pmatrix} = \begin{pmatrix} -3 \\ 4 \end{pmatrix} \leadsto v_B = \begin{pmatrix} -3/5 \\ 4/5 \end{pmatrix}$$

$$v_B \circ v = \begin{pmatrix} 3/5 \\ 4/5 \end{pmatrix} \cdot \begin{pmatrix} -3/5 \\ 4/5 \end{pmatrix} = \frac{-9+16}{25} = \frac{7}{25}$$

$$v_D \circ v = \begin{pmatrix} -3/5 \\ 4/5 \end{pmatrix} \cdot \begin{pmatrix} -3/5 \\ 4/5 \end{pmatrix} = \frac{9+16}{25} = 1$$

- b) Let's assume the following Phong Lighting parameters for the surface:

$$k_{amb} = (0,0,0), k_{diff} = (1,0,0), k_{spec} = (0,1,0), n_{shiny} = 2$$

The parallel light is of intensity  $I_{in} = (1,1,1)$ .

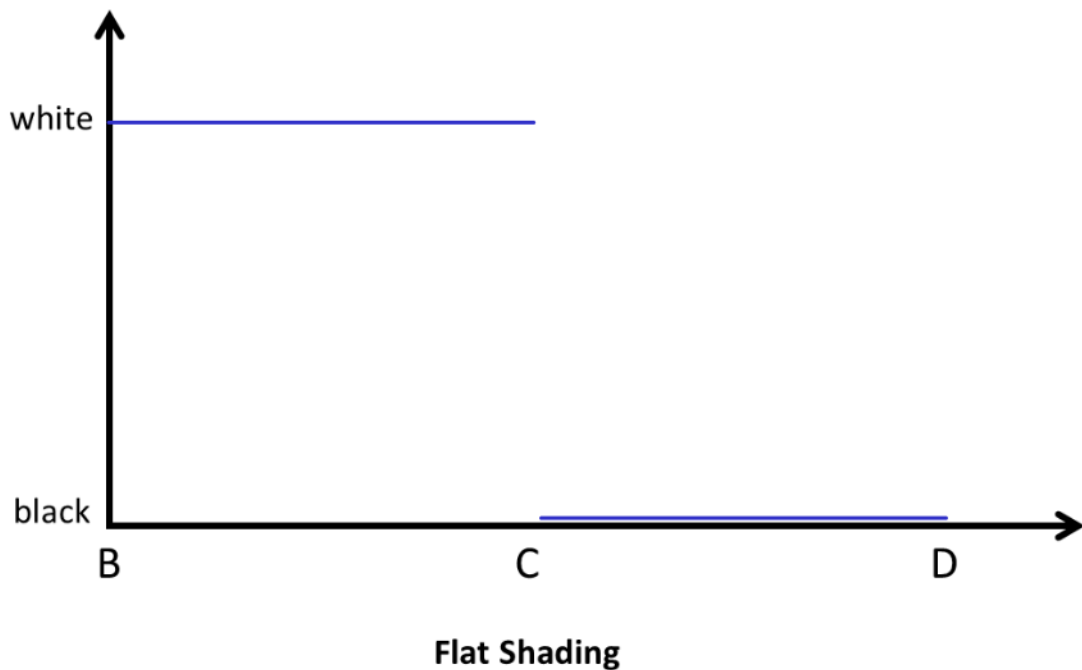
Compute the colors of the two points B and D!

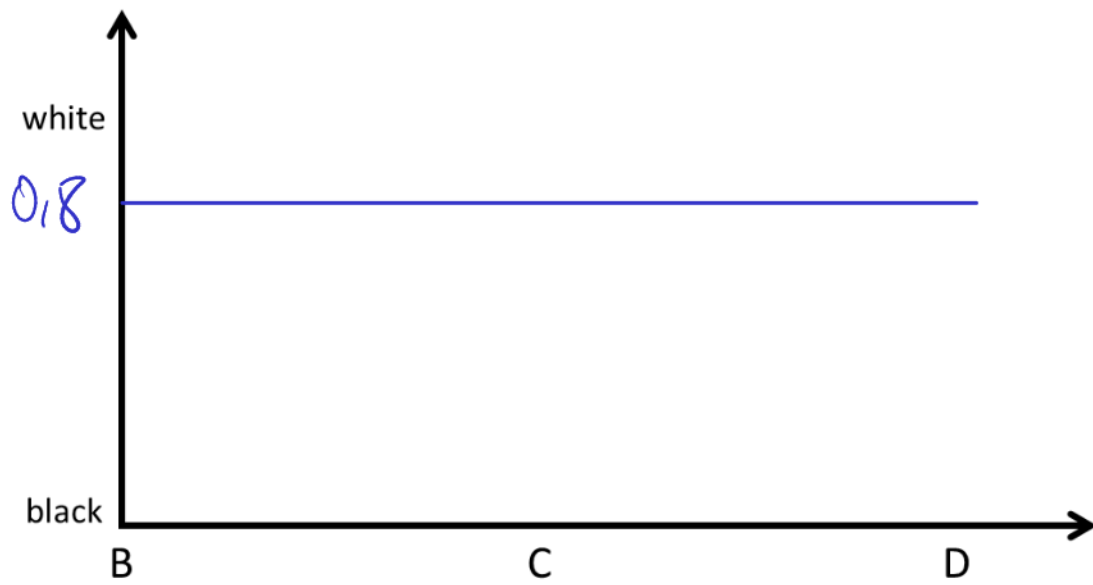
$$\begin{aligned}
 L &= L_{amb} + L_{diff} + L_{spec} \\
 L_{amb} &= \vec{0} \\
 L_{diff} &= \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} 0.8 = \begin{pmatrix} 0.8 \\ 0 \\ 0 \end{pmatrix} \\
 L_{spec} &= k_{spec} I_{in} (v \cdot o)^{n_{shiny}} \\
 L_{spec}_B &= \begin{pmatrix} 0 \\ 7/25^2 \\ 0 \end{pmatrix}, L_{spec}_D = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}
 \end{aligned}$$

$$\begin{aligned}
 L_B &= \begin{pmatrix} 0.8 \\ 0.0784 \\ 0 \end{pmatrix} \\
 L_D &= \begin{pmatrix} 0.8 \\ 1 \\ 0 \end{pmatrix}
 \end{aligned}$$

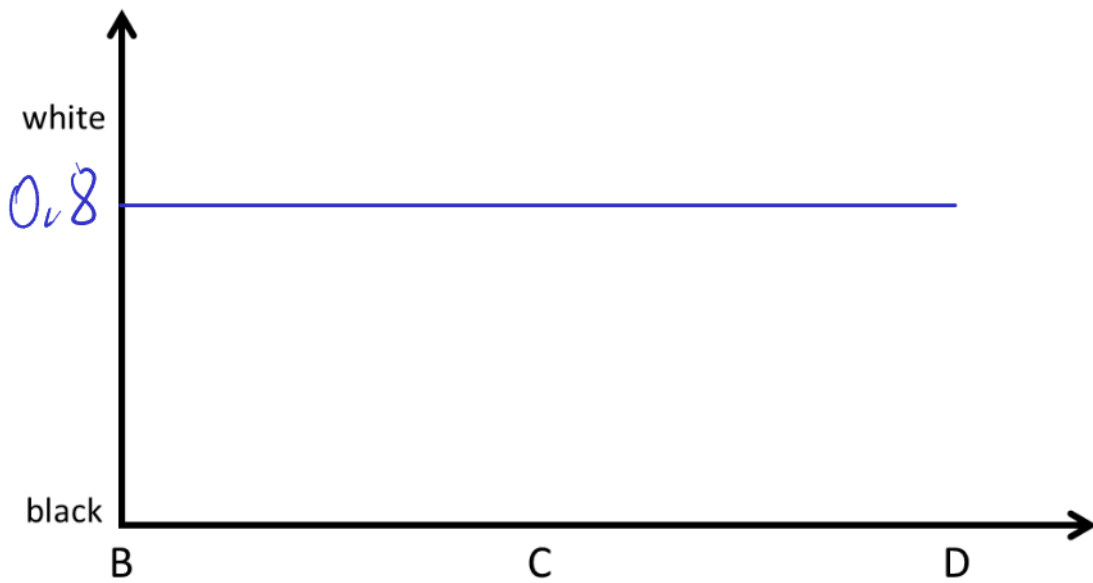
- c) Next, assume that the surface is made of purely **diffuse white material**, so  $k_{amb} = k_{spec} = (0,0,0)$ ,  $k_{diff} = (1,1,1)$ .

Sketch the grey value perceived by the camera along the facets BC and CD, once for **Flat Shading**, once for **Gouraud Shading** and once for **Phong Shading**!





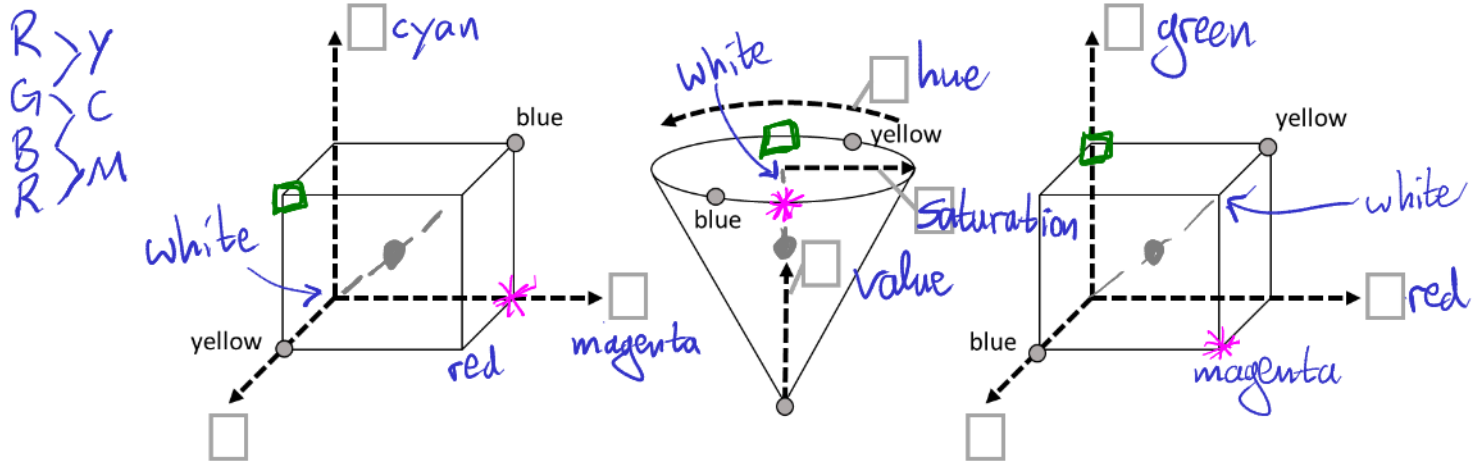
Gouraud Shading



Phong Shading

## Assignment 5: Color Models [10 points]

Below you can see visualizations for the three most common color models, which you know from the lecture. The location of the two colors blue and yellow is given as a reference in all models.



a) For all color models, fill in the names of their axes into the boxes outlined in grey.

b) In all color models, mark the location of the following four additional colors with the given symbol:

- grey
- ✕ white
- ★ magenta
- green

c) For each color model, briefly state what it is most suited for!

RGB: additive → black background  
e.g. computer screen

CMY: subtractive → white background  
e.g. printer

HSV: color picker (perceptual)